Netflix Customer Churn

SQL Analysis Report

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Database	MySQL 8.0
Table	netflix_churn (50,000 rows, 21 columns)
Tools	MySQL Workbench, Python (data generation)
Total Queries	10 (4 Foundation + 6 Advanced)

Business Context

This report presents 10 SQL queries written to analyse customer churn behaviour on a Netflix-style streaming platform. The analysis covers overall churn metrics, segment-level breakdowns, revenue impact, and identification of high-risk user groups. All queries were executed on a MySQL database containing 50,000 user records with 20 behavioural and demographic features.

Key Metrics at a Glance

Metric	Value
Total Users	50,000
Churned Users	15,949
Overall Churn Rate	31.90%
Total Monthly Revenue	$725,620
Revenue at Risk	$239,743 (33%)
Highest Risk Segment	Premium + Low Engagement -- 58.59%
Safest Segment	Basic + High Engagement -- 13.69%

Section 1 — Foundation Queries

These 4 queries establish the baseline metrics — overall churn rate, segment-level breakdowns, and total revenue impact.

Query 1 — Overall Churn Rate

SQL:

```sql
SELECT COUNT(*) AS total_users,
SUM(churned_flag) AS Churned_users,
ROUND(SUM(churned_flag) * 100.0 / COUNT(*),2) as Churn_rate_pct
FROM netflix_churn
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_users	Churned_users	Churn_rate_pct
50000	15949	31.90

Insight:

> **Insight:** Nearly 1 in 3 users have churned. A 31.9% churn rate is significantly above the industry benchmark of 5-8% for streaming platforms, indicating a serious retention problem that requires immediate intervention.

Query 2 — Churn Rate by Subscription Type

SQL:

```sql
SELECT subscription_type, COUNT(*) AS total_users,
SUM(churned_flag) AS churned_users,
ROUND(SUM(churned_flag) * 100.0 / COUNT(*),2) AS Churned_rate_pct
FROM netflix_churn
GROUP BY subscription_type
ORDER BY churned_rate_pct DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

subscription_type	total_users	churned_users	Churned_rate_pct
Premium	12564	4686	37.30
Standard	19988	6497	32.50
Basic	17448	4766	27.32

Insight:

> **Insight:** Premium users churn the most at 37.3% — counter-intuitive but explainable. High-paying users have higher expectations. When content does not meet those expectations or engagement drops, they feel the cost is unjustified and cancel. Basic users churn the least at 27.3%, likely because the lower price point reduces the perceived risk of staying.

Query 3 — Churn Rate by Activity Level

SQL:

```sql
SELECT
    CASE
        WHEN days_since_last_login <= 15 THEN 'Active'
        WHEN days_since_last_login <= 30 THEN 'Moderate'
        ELSE 'Inactive'
    END AS activity_level,
    COUNT(*) AS total_users,
    SUM(churned_flag) AS churned_users,
    ROUND(SUM(churned_flag) * 100.0 / COUNT(*), 2) as churn_rate_pct
FROM netflix_churn
GROUP BY activity_level
ORDER BY churn_rate_pct DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

activity_level	total_users	churned_users	churn_rate_pct
Inactive	24372	12875	52.83
Moderate	12533	2348	18.73
Active	13095	726	5.54

Insight:

> **Insight:** Days since last login is the single strongest churn predictor. Inactive users (30+ days without login) churn at 52.83% — nearly 10x higher than active users at 5.54%. This finding directly informs the recommendation to trigger re-engagement notifications after 20 days of inactivity.

Query 4 — Revenue at Risk

SQL:

```sql
SELECT
    ROUND(SUM(monthly_fee),2) as total_monthly_revenue,
    ROUND(SUM( CASE WHEN churned_flag = 1 THEN monthly_fee ELSE 0 END), 2) AS revenue_at_risk,
    ROUND(SUM( CASE WHEN churned_flag = 1 THEN monthly_fee ELSE 0 END) * 100.0 / SUM(monthly_fee), 2) as revenue_risk_pct
FROM netflix_churn
ORDER BY revenue_risk_pct DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_monthly_revenue	revenue_at_risk	revenue_risk_pct
725620.3	239742.74	33.04

Insight:

> **Insight:** $239,743 in monthly revenue is at risk from churned users — equivalent to $2.87M annually. Even a 10% improvement in retention through targeted interventions would recover approximately $24,000 per month ($288,000 per year).

Section 2 — Advanced Queries

These 6 queries go deeper — identifying high-risk segments, cross-variable patterns, and behavioural profiles that drive targeted retention strategies.

Query 5 — High Risk User Identification

SQL:

```sql
SELECT
    COUNT(*) AS high_risk_users,
    SUM(churned_flag) AS churned_count,
    ROUND(SUM(churned_flag) * 100.0 / COUNT(*), 2) AS churn_rate_pct,
    ROUND(SUM(CASE WHEN churned_flag = 1 THEN monthly_fee ELSE 0 END), 2) AS revenue_at_risk
FROM netflix_churn
WHERE avg_watch_time_minutes < (SELECT AVG(avg_watch_time_minutes) FROM netflix_churn)
AND days_since_last_login > (SELECT AVG(days_since_last_login) FROM netflix_churn);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

high_risk_users	churned_count	churn_rate_pct	revenue_at_risk
12522	8866	70.80	131382.83

Insight:

> **Insight:** 12,522 users (25% of total) are classified as high-risk — low watch time combined with high inactivity. These users churn at 70.8%, putting $131,383 in monthly revenue at immediate risk. This segment should be the primary target for re-engagement campaigns.

Query 6 — Churn Rate by Engagement Level

SQL:

```sql
SELECT
    CASE
        WHEN avg_watch_time_minutes <= 200 THEN 'Low'
        WHEN avg_watch_time_minutes <= 400 THEN 'Medium'
        ELSE 'High'
    END AS engagement_level,
    COUNT(*) AS total_users,
    SUM(churned_flag) AS churned_users,
    ROUND(SUM(churned_flag) * 100.0 / COUNT(*), 2) AS churn_rate_pct
FROM netflix_churn
GROUP BY engagement_level
ORDER BY churn_rate_pct DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

engagement_level	total_users	churned_users	churn_rate_pct
Low	13119	7019	53.50
Medium	14476	4954	34.22
High	22405	3976	17.75

Insight:

> **Insight:** There is a clear 3x difference between low and high engagement churn rates (53.5% vs 17.75%). Watch time is a direct proxy for perceived value — users who watch more, stay more. Improving content discovery and personalised recommendations can shift users from Low to Medium engagement, significantly reducing churn.

Query 7 — Premium High Risk Users

SQL:

```sql
SELECT
    COUNT(*) AS premium_high_risk_users,
    SUM(churned_flag) AS churned_count,
    ROUND(SUM(churned_flag) * 100.0 / COUNT(*), 2) AS churn_rate_pct,
    ROUND(SUM(CASE WHEN churned_flag = 1 THEN monthly_fee ELSE 0 END), 2) AS revenue_at_risk
FROM netflix_churn
WHERE subscription_type = 'Premium'
AND avg_watch_time_minutes < (SELECT AVG(avg_watch_time_minutes) FROM netflix_churn)
AND days_since_last_login > (SELECT AVG(days_since_last_login) FROM netflix_churn);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

premium_high_risk_users	churned_count	churn_rate_pct	revenue_at_risk
3209	2472	77.03	49458.2

Insight:

> **Insight:** 3,209 Premium users are high-risk with a 77% churn rate — the most critical segment. These users pay the highest fee but are the least engaged. At $49,458 monthly revenue at risk, even retaining 20% of them would save nearly $10,000/month. Recommended action: offer a subscription pause option instead of cancellation.

Query 8 — Churn Rate by Customer Tenure

SQL:

```sql
SELECT
    CASE
        WHEN account_age_months <= 6 THEN 'New (0-6m)'
        WHEN account_age_months <= 12 THEN 'Early (6-12m)'
        WHEN account_age_months <= 24 THEN 'Established (1-2yr)'
        ELSE 'Loyal (2yr+)'
    END AS customer_tenure,
    COUNT(*) AS total_users,
    SUM(churned_flag) AS churned_users,
    ROUND(SUM(churned_flag) * 100.0 / COUNT(*), 2) AS churn_rate_pct
FROM netflix_churn
GROUP BY customer_tenure
ORDER BY churn_rate_pct DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_tenure	total_users	churned_users	churn_rate_pct
New (0-6m)	5034	1804	35.84
Early (6-12m)	4979	1591	31.95
Established (1-2yr)	10134	3214	31.72
Loyal (2yr+)	29853	9340	31.29

Insight:

> **Insight:** New users (0-6 months) churn at the highest rate of 35.84%, suggesting a weak onboarding experience. The first 30 days are critical — if users do not find value quickly, they leave. Recommended action: implement a guided onboarding flow with personalised content recommendations during the first month.

Query 9 — Subscription Type vs Engagement Cross Analysis

SQL:

```sql
SELECT
    subscription_type,
    CASE
        WHEN avg_watch_time_minutes <= 200 THEN 'Low'
        WHEN avg_watch_time_minutes <= 400 THEN 'Medium'
        ELSE 'High'
    END AS engagement_level,
    COUNT(*) AS total_users,
    SUM(churned_flag) AS churned_users,
    ROUND(SUM(churned_flag) * 100.0 / COUNT(*), 2) AS churn_rate_pct
FROM netflix_churn
GROUP BY subscription_type, engagement_level
ORDER BY churn_rate_pct DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

subscription_type	engagement_level	total_users	churned_users	churn_rate_pct
Premium	Low	3316	1943	58.59
Standard	Low	5195	2818	54.24
Basic	Low	4608	2258	49.00
Premium	Medium	3690	1514	41.03
Standard	Medium	5757	2001	34.76
Basic	Medium	5029	1439	28.61
Premium	High	5558	1229	22.11
Standard	High	9036	1678	18.57
Basic	High	7811	1069	13.69

Insight:

> **Insight:** Premium + Low Engagement is the most dangerous combination at 58.59% churn. Basic + High Engagement is the safest at 13.69%. This 45-point gap confirms that engagement — not plan type — is the true driver of retention. The platform should focus on increasing engagement across all plan types, particularly for Premium subscribers.

Query 10 — Binge Watchers vs Non-Binge Watchers

SQL:

```sql
SELECT
    CASE
        WHEN binge_watch_sessions > (SELECT AVG(binge_watch_sessions) FROM netflix_churn) THEN 'Binge Watcher'
        ELSE 'Non Binge Watcher'
    END AS user_type,
    COUNT(*) AS total_users,
    SUM(churned_flag) AS churned_users,
    ROUND(SUM(churned_flag) * 100.0 / COUNT(*), 2) AS churn_rate_pct
FROM netflix_churn
GROUP BY user_type
ORDER BY churn_rate_pct DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

user_type	total_users	churned_users	churn_rate_pct
Non Binge Watcher	23446	9726	41.48
Binge Watcher	26554	6223	23.44

Insight:

> **Insight:** Binge watchers churn at nearly half the rate of non-binge watchers (23.44% vs 41.48%). Content that hooks users into watching multiple episodes in one session is a powerful retention tool. Promoting binge-worthy series to moderate users — especially through personalised recommendations — can convert them into loyal watchers.

Summary of Findings

#	Finding	Churn Rate	Recommended Action
1	Inactive users (30+ days)	52.83%	Re-engagement alert at day 20
2	Premium + Low Engagement	58.59%	Subscription pause option
3	High Risk segment overall	70.80%	Priority outreach campaign
4	New users (0-6 months)	35.84%	Improve onboarding flow
5	Non-binge watchers	41.48%	Promote binge-worthy content
6	Basic + High Engagement	13.69%	Protect — reward loyalty